FinTrust Digital Bank - Data Analytics Track

Week 4: Final Project Documentation

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1. Project Summary and Deliverables

This document describes the final Data Analytics solution for FinTrust Digital Bank: a cleaned dataset, SQL analysis, a reproducible Python notebook, a two-page Power BI dashboard, six validated business insights and the testing evidence behind them. The analysis covers 1,500 customers and 12,000 transactions from January to March 2026. (Record counts of 1,501 and 12,001 in earlier weeks include the CSV header row.)

Required deliverable	File	Where
Final Power BI dashboard	FinTrust_Week4_Final_Dashboard.pbix	Section 6
Final SQL scripts	FinTrust_Week4_Final_SQL_Analysis.sql	Section 4
Final Python notebook	Fintrust_Week2_Data_Analysis.ipynb + two CSV snapshots	Section 5
KPI documentation	This document	Section 3
Final business insights and recommendations	FinTrust_Week4_PartD_Final_Business_Insights.pdf	Section 7
Validation evidence	PartE PDF, FinTrust_Week4_Validation_Checks.py, FinTrust_Week4_Validation_Output.txt	Section 8
Data quality documentation	This document + FinTrust_Week2_DataQuality_Cleaning_Workbook.xlsx	Section 2
Review of Weeks 1-3	FinTrust_Week4_PartA_Final_Review.pdf	Section 8
SQL data repair (import issue)	FinTrust_SQL_Amount_Repair.sql	Sections 2, 8
Final project documentation	This document	All

2. Data and Data Quality

Sources. FinTrust_Customer_Data (1,500 customers, 12 columns) and FinTrust_Transaction_Data (12,000 transactions, 11 columns). Both are synthetic datasets supplied by AnalystLab Africa. Key fields: Customer_Segment (Everyday, Premium, Student, SME), Account_Status, Amount_NGN, Channel (Mobile App, POS, Web, ATM, USSD), Transaction_Type, Transaction_Status, Risk_Review_Flag and International_Transaction.

Check	Result	Action
Missing values	96 Device_Type and 96 Location values missing in Transaction_Data (0.8% each)	Imputed as "Unknown" in the cleaned dataset
Duplicates	No duplicate Customer_ID or Transaction_ID	None needed
Invalid values	No negative amounts; no implausible ages	None needed

Check	Result	Action
Amount range	NGN 100.17 to NGN 693,454.45, mean NGN 46,706.45, right-skewed	Kept all values; skew noted in analysis
Data types	SQL Server read Risk_Review_Flag and International_Transaction as bit (1/0); raw CSV and Power BI hold Yes/No; pyodbc returns True/False	Each tool handled with the matching comparison (documented in Section 8)
Import reconciliation (Week 4)	SUM(Amount_NGN) in SQL Server was 554,157,109.34 vs 560,477,354.85 in the CSV. 128 whole-number amounts (total NGN 6,320,246) were imported as NULL, and the column was stored as float	Repair script restores the 128 amounts and converts the column to DECIMAL(18,2); counts and rates were never affected
Re-verification (Week 4)	All checks re-run on the source files with identical results	No drift

3. KPI Documentation

KPI	Definition	Final value	Purpose
Total Customers	Distinct count of Customer_ID	1,500	Size of the customer base
Total Transactions	Count of Transaction_ID	12,000	Platform activity volume
Total Transaction Value	SUM(Amount_NGN)	NGN 560.48M	Value flowing through FinTrust
Average Transaction Value	SUM(Amount_NGN) / COUNT(Transaction_ID)	NGN 46.71K	Typical transaction size
Transaction Success Rate	Successful transactions / all transactions	90.47%	Reliability
Risk Review Rate	Flagged transactions / all transactions	19.60%	Share sent for manual risk review
Average Daily Value	SUM(Amount_NGN) / DISTINCTCOUNT(transaction date)	NGN 6.08M (Jan), 6.28M (Feb), 6.33M (Mar)	Fair trend comparison across months of different length

Rates in Power BI use the COUNTROWS / FILTER / DIVIDE pattern, because the flag fields are text. All six headline KPIs were recomputed from the raw CSVs in Python and match the dashboard cards. After the import repair described in Section 2, SQL Server returns NGN 560,477,355.91 with 0 NULL amounts, within NGN 1.06 of the CSV total.

4. Final SQL Analysis (Part B)

The final script contains eight queries selected from the 16 written in Weeks 2 and 3. Together they use JOIN, GROUP BY, CASE, aggregations, a subquery, CTEs and window functions (SUM() OVER, RANK, NTILE). Each query in the file carries its result and a plain-language explanation.

Query	Techniques	What it reveals
1. Segment value and share	JOIN, GROUP BY, window	Everyday holds 46.65% of value, in line with its 47.4% of customers
2. International vs domestic risk	CASE, GROUP BY	36.88% vs 18.88% flagged
3. International status x channel	CASE, GROUP BY	International roughly doubles risk in every channel group; no channel stands out
4. Failure/reversal by channel and type	GROUP BY, HAVING, CASE	Highest cells (10%) are within normal variation of the 7.97% average
5. Customer spend ranking	CTE, RANK()	Top spender NGN 1.71M; wide spread in spend and frequency
6. Top 1% transactions	CTE, NTILE()	Top 1% flagged 35.83% vs 19.44% for the rest
7. Above-average spenders by segment	Subquery, JOIN	605 of 1,500 customers; similar share (37-46%) in every segment
8. Segment x transaction type	JOIN, CASE, GROUP BY	Transfer is the largest type by value in every segment

5. Python Analysis (Part A, notebook)

The notebook contains ten visualisations (five from Week 2, five from Week 3): customer segments, amount distribution, channel volume, risk rate by international status, status mix, day-of-week volume, risk heatmap (channel x international), spend vs frequency, transaction type by segment, and success rate by value range. **Reproducibility:** the first cell loads both tables from SQL Server through pyodbc and saves CSV snapshots; if no database is available, it loads the snapshots instead, so every later cell runs on any machine. The notebook is run top to bottom (Restart and Run All) before submission, after the SQL repair so the snapshots hold the correct amounts.

6. Final Power BI Dashboard (Part C)

Page	Content
Overview	Six KPI cards; customer segments; transaction value by type (donut); transactions by channel; monthly transaction value (column chart, zero-based axis); transaction status distribution (percentage bars); risk review by channel (100% stacked); Key Highlights
Transaction and Risk	Volume by day of week; risk rate matrix (channel x domestic/international); customer spend vs frequency; success rate by value range; Key Highlights

Review performed: visual clarity, KPI accuracy, slicers (status, type, segment, risk flag, channel, date), labels, titles, consistency and user experience. Changes made in Week 4: highlights rewritten to match validated findings; trend chart changed to a zero-based column chart titled to show February has 28 days; status pie replaced with a percentage bar chart because small slices hid their labels; risk-by-channel changed to a 100% stacked bar; matrix columns renamed Domestic / International; axis titles, slicer titles and page tabs standardised.

7. Final Business Insights (Part D)

Insight	Key evidence	Recommended action
International transactions carry about double the risk-review rate	36.9% vs 18.9%; p < 0.001	Dedicated review rule for all international transactions
Risk-review rate rises with transaction value	18% below NGN 50K, 41% above NGN 250K	Keep value-based thresholds; report flag rate by value tier
About 8% of transactions fail or reverse in every channel	ATM 6.9% to USSD 8.7%; p = 0.07	Platform-wide failure-cause review; add a failure-reason field
Segments differ in size, not behaviour	Spend per customer NGN 365K to 388K; risk 19.3% to 20.3%	Build behaviour-based groups for targeting
Mobile App and Transfer carry the business	42.5% of transactions; 42.4% of value	Prioritise their reliability and support
February dip is a calendar effect	133 transactions per day in all three months	Use average daily value as the trend KPI

Full Finding, Evidence, Business Meaning, Recommended Action tables are in the Part D document.

8. Testing, Validation and Corrections Log (Parts A and E)

Validation. Eight findings were validated by recomputation from raw data and significance tests (chi-square, Kruskal-Wallis, binomial, 95% intervals). Results: five confirmed or refined, three revised. The full evidence is in the Part E document, and the script and output file let anyone repeat the checks.

Week	Issue found	Correction
2	SQL Server typed the flag fields as bit, so Yes/No comparisons failed	All SQL compares to 1/0
2	pyodbc returned True/False, so charts of risk by international status were wrong	Mapped True/False in Python; results re-matched to SQL
3	Risk heatmap returned NaN values	Mapping corrected to True/False; heatmap re-verified
3	Risk Review Rate used AVERAGE() on a text field	Rebuilt with COUNTROWS / FILTER / DIVIDE
3	A scatter plot mix-up was found during testing	Corrected and re-verified against the spend and frequency results
4	ATM + International (46.0%) presented as the top risk hotspot	Only 63 transactions; revised to international status overall as the driver
4	Failures described as concentrated in USSD	Not significant (p = 0.07); revised to a platform-wide rate of about 8%
4	February dip treated as a possible real fluctuation	Calendar effect (28 days); Average Daily Value added

Week	Issue found	Correction
4	SQL Query 3 stated that risk factors compound	International effect is about 1.95x in both channel groups; comment rewritten
4	SQL Server total (NGN 554.16M) did not match the CSV (NGN 560.48M); 128 whole-number amounts imported as NULL; column stored as float	Found by reconciliation; repaired with FinTrust_SQL_Amount_Repair.sql; Week 2 SQL value totals (about 1.1% low) noted rather than edited
4	SQL Query 1 comment said Everyday value exceeds the other segments combined, and its listed values were affected by the NULL amounts	Comment corrected; values replaced with raw-CSV figures (Everyday NGN 261.46M, 46.65%)
4	Dashboard highlights referred to "January's dip", USSD as a gap, and 88-92% success	Rewritten to match validated findings (89-92%)
4	Notebook not portable (live SQL Server dependency)	CSV snapshot fallback added to the data-loading cell

9. Limitations

- The data covers three months, so seasonality, long-term growth and recurring cycles cannot be assessed.

- The datasets are synthetic; results show what the analysis method finds, not real FinTrust behaviour.

- Risk_Review_Flag records transactions sent for review, not confirmed fraud, so risk findings describe review activity rather than loss.

- There is no failure-reason field, so the analysis shows how often transactions fail but not why.

- International transactions number 480, so any breakdown within them (for example by channel) has wide uncertainty.

- Week 2 SQL value totals were about 1.1% low because of the import issue (counts and rates were unaffected); those files were left as submitted.

- The analysis shows association, not cause. For example, it cannot say that value or international status causes a flag.

- Dashboard figures reflect the loaded data snapshot and do not refresh from a live source.

10. Integration with the Wider FinTrust Solution

Output	How other parts of the solution can use it
KPI definitions (Section 3)	A shared set of metric definitions so every track reports the same numbers
Risk patterns (international status, transaction value)	Candidate input features for any risk or anomaly model built by the data science or machine learning tracks; the flag is a review outcome, so it should be treated with care as a training label
Insights and validated findings	Source facts for any support assistant or reporting feature, with the stated limitations

Output	How other parts of the solution can use it
Final SQL and notebook	Reusable queries and CSV snapshots that other team members can run without my database
Dashboard	A management view of customers, transactions, channels, status and risk, ready to extend with new data

Assumption: the table above describes how my outputs can be used. It does not describe what other tracks have actually built.

11. Lessons Learned

A finding is only as strong as the effort to break it: testing my own conclusions made them smaller but more honest. Data types and definitions matter, as the bit/Yes-No issue and the 28-day month both showed. A chart is only useful when it states what it means for the business.